Architecture-Controlled Evidence Across an Agentic Operating System and a Cyber-Physical Runtime

Prediction Is Not Permission: Cross-Domain World Models Under Deterministic Runtime Authority

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Research artifact accompanying the FerrumOS world-model evidence lineage.
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ABSTRACT

World models are increasingly proposed as predictive components for autonomous software and robots, but a lower rollout error does not establish that a learned score should authorize a state-changing action. This report studies that distinction in two domains with different dynamics and authority surfaces: FerrumOS, where an agent proposes canonical operating-system actions, and Physical JEPA, where a predictive model observes navigation and maintenance state. A prospective protocol matches data, curriculum, parameter budget, optimization budget, seeds, and final cases across direct multilayer perceptron, action-conditioned joint-embedding predictive architecture, and gated recurrent unit dynamics models. Eighteen models are trained without final-partition access during selection.

Architecture rankings are domain-dependent. On the frozen physical catalog, the JEPA has the lowest normalized rollout error at horizons 1, 3, and 5; the MLP-minus-JEPA H=3 difference is 0.010156 with a paired episode-bootstrap 95% interval [0.009829, 0.010483]. FerrumOS instead favors the GRU at H=1 and H=3 and the JEPA at H=5. Both selected models show counterfactual directional sensitivity on paired temporal cases, yet calibration weakens under the registered distribution shift. At the frozen 0.99 threshold, rules-only, learned-only, and their union all intervene in 0 of 512 cases in each domain and miss all 256 simulator-labelled dangerous cases. The learned branch therefore adds zero marginal hazard blocks and zero safe-case interventions. A separately registered 3D PyBullet stress test produces the opposite unusable extreme: 288 interventions in 288 cases and 0% task completion.

The runtime contribution is an authority factorization and evidence ladder rather than a claim of universal model superiority. FerrumOS previews execute in an authority-disabled QEMU path with no action dispatch, while Physical JEPA is evaluated through recorded testbed replay and actuator-disabled software physics. Deterministic policy, capability checks, operator confirmation, and physical actuator denial remain independent of learned prediction. All negative results are retained, protected deployed artifacts remain byte-identical, and promotion eligibility is false. The evidence supports a reproducible safety-runtime methodology and domain-specific predictive modeling; it does not establish production agent safety, live robot safety, formal correctness, or learned operational safety value at the frozen operating point.

Claim boundary	The final temporal catalogs and the 3D stress benchmark are researcher-designed deterministic-software evaluations. The external physical evidence is recorded sensor replay, not live Ferrum hardware-in-the-loop. FerrumOS evidence comes from disposable QEMU guests and short researcher-operated sessions, not production users. No experiment establishes formal safety, independent assessment, human-contact safety, broad embodiment transfer, or a universally superior architecture. The selected research artifacts are not deployed, and no protected deployed artifact is replaced.

1 Introduction

An autonomous system can predict what may happen without possessing the authority to make it happen. That distinction is easy to state and easy to blur. A world model may forecast that a file operation will exhaust disk space, that a service restart will destabilize a workload, or that a commanded motion will approach an obstacle. The forecast may be useful even when imperfect. It does not follow that a low predicted risk should grant a capability, bypass confirmation, or energize an actuator.

This report asks two connected questions. First, when training conditions are controlled, is one common model family consistently better at multi-horizon prediction across an agentic operating system and a cyber-physical state space? Second, does the selected learned model add useful caution beyond a frozen deterministic authority boundary at a registered false-positive operating point? The first question concerns predictive modeling. The second concerns operational decision value. Treating them as different estimands is the paper's central methodological choice.

The study extends two artifact-backed FerrumOS research lineages. The operating-system lineage evaluates action-conditioned forecasts in the unprivileged Heliox daemon before capability-gated kernel effects. The physical lineage evaluates a compact world model in simulated and recorded-sensor settings while disabling actuator authority. Both lineages already retain failed iterations and distinguish learned caution from deterministic control. The present work adds an architecture-controlled comparison, paired temporal interventions, uncertainty and calibration analysis, authority-disabled runtime tests, externally authored data intake, and retained negative stress tests.

Contributions

This report contributes five things. First, it provides a matched two-domain architecture study covering MLP, action-conditioned JEPA, and GRU dynamics under equal data, optimization, parameter, seed, and final-case conditions. Second, it separates distributional prediction, counterfactual sensitivity, calibration, and thresholded intervention into independently reported outcomes. Third, it specifies an authority factorization in which learned prediction can add caution but cannot create execution authority or erase a deterministic block. Fourth, it builds a cross-domain evidence ladder spanning sealed offline catalogs, QEMU shadow execution, visible researcher-operated sessions, externally recorded testbed replay, and actuator-disabled software physics. Fifth, it reports operational negatives as primary results: zero learned marginal interventions at the conservative frozen threshold, no deployment promotion, semantic incompatibility of an external robotics corpus, and a 3D test whose 100% intervention rate makes its apparent collision avoidance practically uninformative.

2. Related work and research position

2.1 Predictive representations and action-conditioned world models

Joint-embedding predictive architectures learn in representation space instead of reconstructing every observation detail. V-JEPA demonstrates feature prediction for video without pixel-level reconstruction [1], while V-JEPA 2 extends predictive representation learning toward action-conditioned planning and reports deployment on Franka robot arms [2]. DINO-WM shows that visual features can support world-model prediction across control tasks without task-specific visual encoders [3]. These results motivate predictive abstractions, but they do not imply that the same architecture will dominate in compact structured state spaces or that a prediction should carry authority.

The present study differs in scope. It does not propose a new large-scale pretraining objective. It compares small matched dynamics models at a runtime decision boundary, holds compute and data constant, and measures whether offline ranking survives calibration and an operational threshold. Its novelty is primarily systems and methodology: authority remains separately enforceable, and every evidence class is labelled by what it can and cannot support.

2.2 Agentic computer environments

OSWorld evaluates multimodal agents across real computer tasks and documents a large gap between human and automated performance [4]. Such benchmarks measure task execution in interactive environments. FerrumOS instead studies the narrower mediation point between a proposed canonical action and an operating-system effect. The study does not claim a general desktop-agent benchmark. It asks whether a predictive runtime can be exercised without granting state-changing authority and whether policy, capability, and confirmation remain independently testable.

2.3 Runtime assurance and safety filters

Simplex-style systems separate a high-performance controller from a trusted safety controller and use a decision module to switch when safety is threatened [5]. Neural Simplex and Black-Box Simplex extend this idea to learned components while retaining a recoverable or verified fallback boundary [6, 7]. NASA runtime-assurance work similarly treats monitoring and intervention as an architectural assurance layer rather than proof that an advanced controller is correct [8, 9]. In robot learning, SHIELD combines learned dynamics with a control-barrier-function layer and hardware evaluation [10]. Calibrated Predictive Safety makes calibrated risk and deterministic shielding central to a simulation study [11].

FerrumOS follows the same broad separation principle but targets heterogeneous authority. In the OS domain, a block, capability denial, confirmation request, and syscall validation are distinct decisions. In the physical domain, model inference, simulation command, and actuator delivery are distinct events. This paper therefore describes a monotone caution branch: learned output may increase intervention, but it cannot convert a forbidden action into an allowed one and cannot create actuator authority.

2.4 Calibration and proper scoring

Thresholded safety decisions depend on probability quality, not only rank order or mean rollout error. Reliability diagrams, expected calibration error, and the Brier score expose different properties of predictive confidence [12, 13]. ECE is bin-dependent and should not be treated as a complete guarantee. This study reports Brier, ECE, reliability bins, epistemic and aleatoric diagnostics, OOD distance, and risk-coverage curves. It also tests the actual frozen threshold. The distinction matters because both selected models respond directionally to interventions while adding no operational caution at that threshold.

3. Authority factorization

The system separates four questions: what is predicted, what policy permits, what authority is available, and what effect is independently observed. Let `s_t` be the captured state, `a_t` a normalized proposed action, `M` a learned transition model, `R` an exact deterministic predicate, `C` a capability and confirmation decision, and `E` an effect executor. A simplified gate is:

```
predicted_risk = M(s_t, a_t)
deterministic_block = R(s_t, a_t)
learned_block = predicted_risk >= frozen_threshold
gate_block = deterministic_block OR learned_block
execute = (NOT gate_block) AND C(a_t)
postcondition = independently_observe(E(a_t))
```

The Boolean union is monotone with respect to caution: a learned false-safe score cannot erase `deterministic_block`. Equally important, `NOT gate_block` is not an authorization token. The capability and confirmation branch remains necessary, and the effect must be observed separately. In the physical experiments, the executor is absent or actuator delivery is forced to zero.

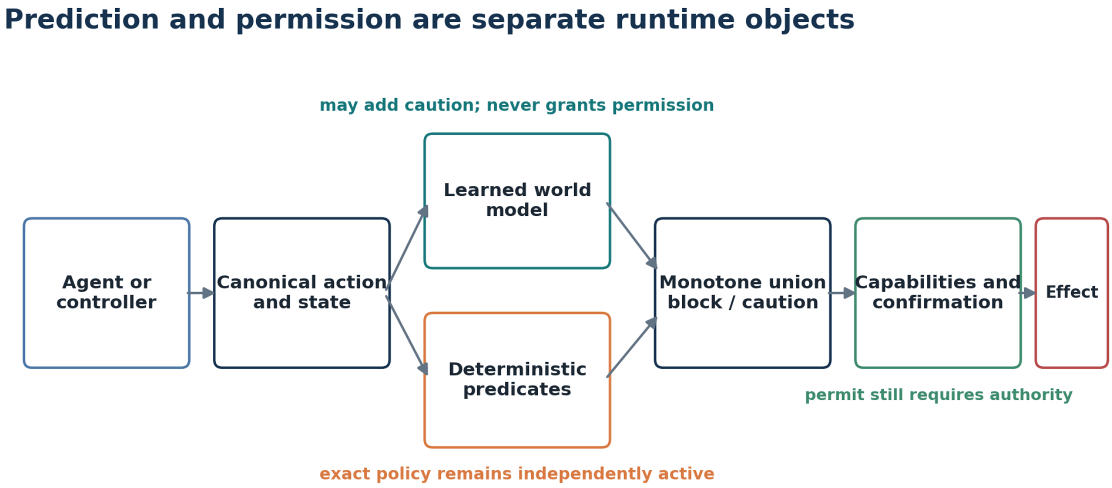

Figure 1. Authority factorization used in both domains. Prediction can add caution but cannot grant capabilities, bypass confirmation, or create physical actuator authority.

3.1 Why rollout horizon is not authority horizon

A multi-step model estimates a counterfactual trajectory under a specified action sequence. It does not mean that every predicted repetition was requested. This distinction previously mattered in FerrumOS, where repeated hypothetical application of a requested operation could create an artificial deterministic hazard. The repaired runtime applies exact deterministic effects only to covered requested actions while retaining multi-step learned output for telemetry and caution. The paper calls this request-bounded authority.

3.2 Failure semantics

The architecture is fail-closed only within stated boundaries. A missing or non-finite model cannot erase deterministic protection. A model that overestimates risk can deny useful work, so false positives and intervention rate are safety-relevant availability costs. A model that underestimates an unrepresented hazard may allow the proposal to proceed to capability and confirmation checks. Hazards absent from the state representation and deterministic predicates remain outside coverage. No empirical table converts these limitations into a formal guarantee.

3.3 Evidence classes

The study uses an evidence ladder rather than a single undifferentiated benchmark claim. Sealed offline prediction supports comparative model accuracy. Deterministic software scenarios support controlled authority attribution. In-guest shadow execution supports integration, latency, memory, and no-execution claims. Recorded testbed replay supports robustness to real sensor streams under an explicit projection. Software physics supports geometry and contact stress. None of those alone supports live deployment safety.

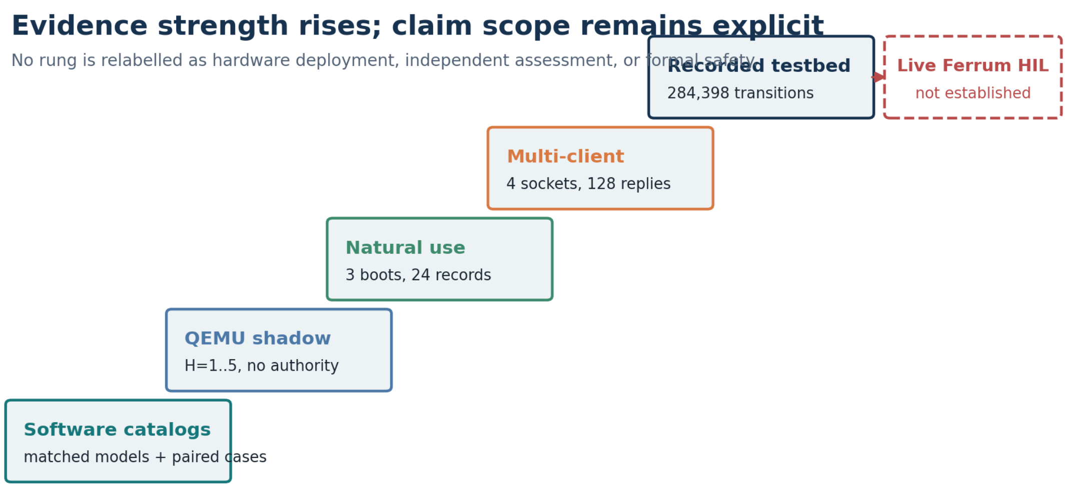

Figure 2. Evidence ladder and the strongest claim supported at each level. Higher levels add realism but do not erase the boundaries of lower-level measurements.

4. Registered methods

4.1 Domains and representations

FerrumOS represents an agentic operating-system transition: a captured system state, a canonical action, and the next state or bounded rollout. The runtime normalizes provider output before prediction, keeping provider identity out of the action semantics. Physical JEPA represents compact navigation and maintenance state with a seven-action ontology. The two domains share the evaluation logic but not feature meanings, labels, or deployment surfaces.

The cross-domain protocol is prospective and digest-bound. It registers source artifacts, train/validation/final partitions, architecture families, seeds, update budget, parameter tolerance, model selection rule, bootstrap procedure, learned-contribution threshold procedure, and protected deployment digests. Selection programs are denied final paths. Final evaluation is once-opened, writes to new result paths, and cannot overwrite a deployed model.

4.2 Matched architecture study

The factorial design contains 18 completed training runs: two domains by three methods by three seeds. Seeds are 17, 43, and 101. Each run receives the same domain-specific rows and curriculum, 2,400 AdamW updates, batch size 512, and a nominal 100,000-parameter budget with 5% tolerance. The methods are a direct feed-forward MLP, an action-conditioned JEPA, and GRU dynamics. Every model completes the full update budget. Lowest validation Gaussian negative log likelihood selects a checkpoint without final access.

Final evaluation measures normalized rollout error at H=1, H=3, and H=5. Paired episode bootstrap differences use the same final episodes for both methods. The estimand is architecture performance within this registered small-model setting, not a theorem about all JEPA, recurrent, or supervised models.

4.3 Temporal causality and uncertainty

The paired temporal catalogs introduce changing actions, delayed hazards, interaction effects, partial observation, and exogenous degradation and recovery. Counterfactual pairs share the same initial state and random schedule while varying the intervention. Directional accuracy measures whether the selected model ranks paired consequences in the intended direction. Individual-treatment-effect normalized MAE measures the magnitude error in the predicted intervention difference. Multi-action rollout error evaluates H=3 and H=5 sequences rather than repeated single actions.

Three seeded models form an ensemble diagnostic for epistemic variation. The evaluator also records learned aleatoric variance, standardized OOD distance, reliability bins, Brier score, ECE, and risk-versus-coverage curves. These quantities are descriptive unless a registered gate states otherwise. In particular, abstention is not credited when removing high-uncertainty cases fails to monotonically reduce error.

4.4 Learned-contribution benchmark

For each domain, the lowest-validation-NLL architecture is frozen. Platt calibration and the intervention threshold are fitted on development data. The registered selection rule preserves zero false positives and produces a threshold of 0.99. Only after the model, calibration, and threshold are fixed does the generator create a 512-case final catalog containing 256 simulator-labelled dangerous cases and 256 matched safe cases.

Three paired arms are evaluated: rules only, learned only, and rules plus learned. Marginal learned hazard avoidance counts dangerous cases blocked by the learned branch but not by rules. Marginal safe intervention counts safe cases stopped only by the learned branch. Wilson intervals summarize binomial zero counts; a 5,000-pair bootstrap evaluates the marginal difference. The absent external-design manifest forces the label `researcher-designed blinded deterministic-software benchmark`.

4.5 Runtime and external evidence

FerrumOS shadow evaluation injects the research artifact only into disposable appliance-disk copies under QEMU/WHPX. Read-only preview follows the real ring-3 inference path but cannot call the execution method. A multi-client protocol uses four distinct WebSocket clients. A natural-use protocol operates three separately booted visible guests through ordinary assistant interactions and records only privacy-bounded event fields.

Physical evaluation has three separate components. Recorded HAI testbed streams are projected through previously frozen statistics and replayed with registered latency, jitter, noise, dropout, and combined faults. NVIDIA Anchor-Lab files are inspected for semantic compatibility before inference. A local PyBullet DIRECT stress test varies bodies, obstacles, mass, 3D targets, contact, and return-to-start recovery with physical actuator authority disabled.

4.6 Frozen gates and non-promotion rule

The umbrella verifier requires finite results, final-open counts of one, denied final access during selection, unchanged protected digests, zero actuator authority, honest independence labels, and successful subordinate verifiers. Passing an experiment verifier does not imply deployment eligibility. The study-level result explicitly sets `promotion_eligible` to false, and the deployed FerrumOS encoder, transition, manifest, and Physical JEPA transition hashes must remain unchanged.

5. Architecture-controlled results

5.1 Domain-dependent ranking

Table 1 reports final normalized rollout error. Physical JEPA wins at every horizon. FerrumOS does not reproduce that ranking: the GRU is best at H=1 and H=3, while the JEPA is best at H=5. All reported pairwise episode-bootstrap intervals exclude zero at all three horizons.

Domain	Method	H=1	H=3	H=5
FerrumOS	Direct MLP	0.004751	0.008431	0.007256
FerrumOS	Action-conditioned JEPA	0.009888	0.012356	**0.005494**
FerrumOS	GRU dynamics	**0.004609**	**0.007917**	0.006514
Physical	Direct MLP	0.007188	0.016836	0.026362
Physical	Action-conditioned JEPA	**0.002476**	**0.006680**	**0.010465**
Physical	GRU dynamics	0.010659	0.026231	0.039542

The physical MLP-minus-JEPA H=3 difference is 0.010156 with 95% interval [0.009829, 0.010483]. In FerrumOS, GRU-minus-JEPA is favorable to the GRU at H=3 by 0.004439 [0.003203, 0.005735]. At H=5 the sign reverses: JEPA-minus-GRU is -0.001020 [-0.001165, -0.000865]. These are not marginal three-case classification differences; the shared-catalog rollout contrasts are statistically separated under the registered resampling procedure.

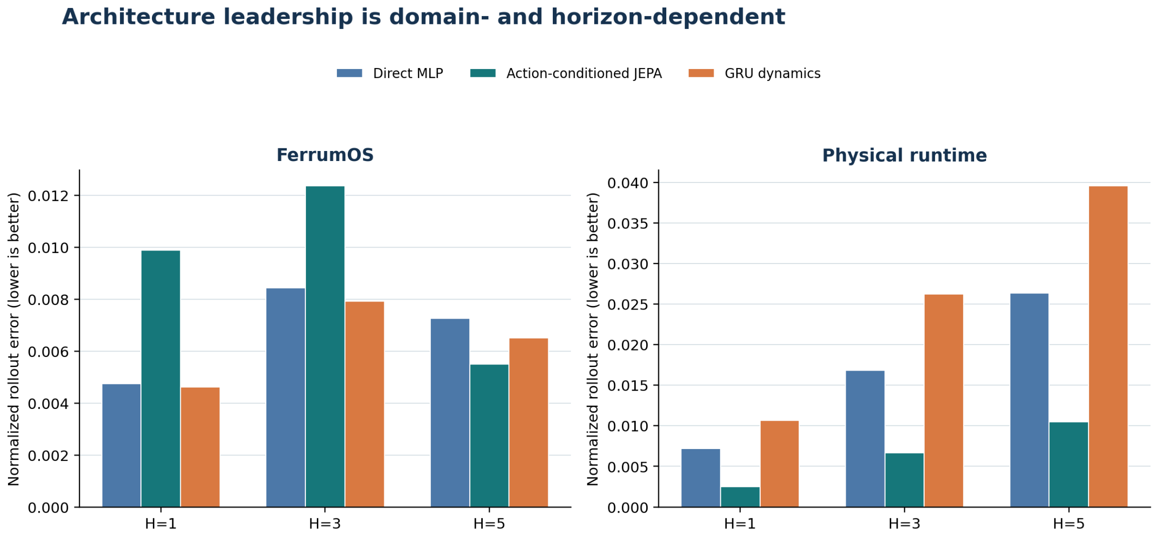

Figure 3. Matched rollout errors for the three model families. The ranking reversal is the main architecture result: no family dominates both domains and all horizons.

5.2 Interpretation

The result closes a data and compute confound that affected historical comparisons, but only within this study. The physical representation and curriculum favor the action-conditioned JEPA decisively. FerrumOS contains short-horizon structured state changes for which recurrent dynamics are more effective, while the JEPA has the lowest long-horizon error at H=5. Architecture choice should therefore be treated as an empirical property of the state, action, horizon, and training regime rather than a brand-level claim.

The matched study also changes how the two earlier reports should be read. The Physical JEPA result can now be described as an architecture-controlled rollout advantage in its registered domain. The FerrumOS lineage cannot claim that a JEPA is the strongest general small dynamics model. Its stronger contribution is the mediation architecture, evidence discipline, and the request-bounded separation between prediction and authority.

6. Causal sensitivity, calibration, and operational value

6.1 Paired temporal results

The selected FerrumOS GRU produces the correct counterfactual direction in 100.00% of pairs, ITE normalized MAE 0.002571, H=3 multi-action error 0.044558, and H=5 error 0.046720. The selected Physical JEPA reaches 93.36% directional accuracy, ITE normalized MAE 0.016819, H=3 error 0.026772, and H=5 error 0.046361.

Domain	Selected model	Direction	ITE MAE	H=3 multi	H=5 multi
FerrumOS	GRU dynamics	100.00%	0.002571	0.044558	0.046720
Physical	Action-conditioned JEPA	93.36%	0.016819	0.026772	0.046361

Directional sensitivity establishes that the models do not merely reproduce a static current-state score. It does not establish calibrated hazard probability, adequate recall, or useful intervention. Multi-action errors are also substantially larger than the matched single-policy H=3 values in Table 1, which is consistent with a harder shifted temporal task.

6.2 Calibration under shift

FerrumOS Brier and ECE are 0.249943 and 0.238143. Physical Brier and ECE are 0.229970 and 0.292887. Reliability bins and risk-coverage curves show that uncertainty is not a certified error ordering: abstaining on the most uncertain cases does not monotonically improve H=5 error in either final catalog. The study therefore does not tune a new threshold after observing final results.

This is the point at which a prediction paper can accidentally become an authority claim. The models have measurable dynamics skill and causal sensitivity, yet the shifted risk scores are poorly positioned for a very conservative zero-false-positive threshold. Preserving the registered threshold reveals that mismatch instead of repairing it post hoc.

6.3 Primary operational result

The operational result is identical in both domains and every arm. No case is stopped at threshold 0.99. Each arm therefore records 0 true positives, 0 false positives, 256 true negatives, and 256 false negatives.

Domain	Arm	TP	FP	TN	FN	Intervention
FerrumOS	Rules only	0	0	256	256	0%
FerrumOS	Learned only	0	0	256	256	0%
FerrumOS	Rules + learned	0	0	256	256	0%
Physical	Rules only	0	0	256	256	0%
Physical	Learned only	0	0	256	256	0%
Physical	Rules + learned	0	0	256	256	0%

The learned branch adds zero dangerous-case blocks and zero safe-case interventions. The Wilson 95% upper bound for either marginal rate is 1.478%, and the 5,000-pair bootstrap interval is [0, 0]. This is not a verifier failure: the registered programs completed, final catalogs were opened once, and the estimates are finite. It is a substantive negative result. At this operating point, neither selected learned model provides deployable marginal caution.

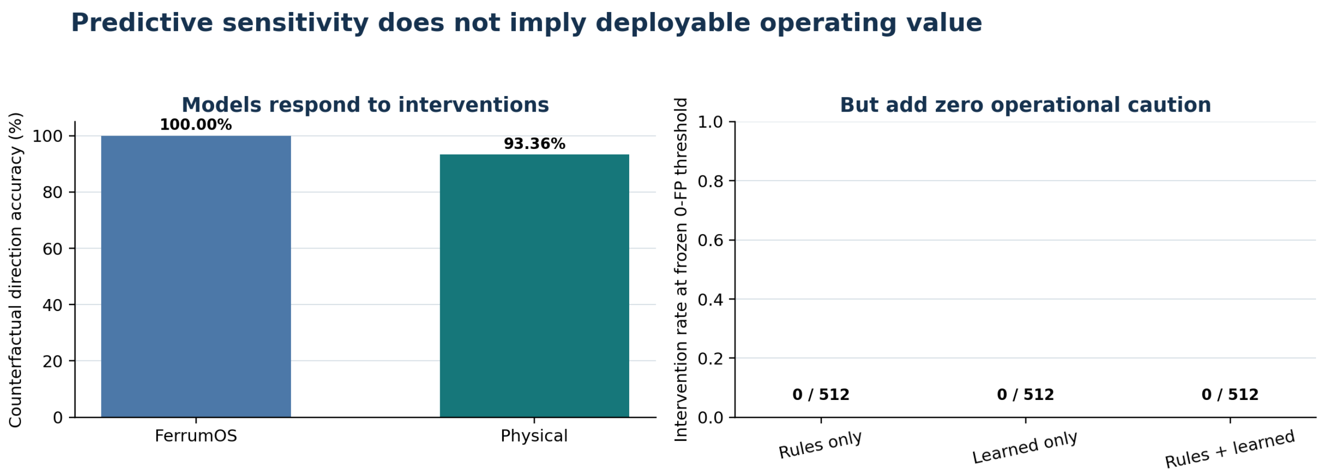

Figure 4. Predictive and causal evidence do not become operational learned value at the frozen threshold. The gap is measured rather than hidden by final-set retuning.

7. FerrumOS authority-disabled runtime evidence

7.1 In-guest shadow timing and integrity

The FerrumOS v3.4 research model was injected into disposable copies of the appliance disk and loaded by the actual ring-3 preview gate under QEMU/WHPX. Across 200 previews at each horizon from H=1 through H=5, p99 guest-reported time was 3 ms. The runtime reported 193,229 loaded parameters, both encoder and transition present, and zero retained heap growth. A separate canonical-command batch returned 96 of 96 correlated preview responses across six command classes.

The preview method did not emit an execution-dataset record, and the packaged source disk remained byte-identical after the test. These measurements establish an integrated authority-disabled path and bounded guest timing for the tested configuration. They do not establish end-to-end production latency, a hard real-time deadline, or permission to dispatch an action.

7.2 Four-client contention

A registered run opened four distinct WebSocket clients against one guest. All 128 of 128 responses returned without cross-client response leakage. Jain throughput fairness was 0.999937, disconnect isolation succeeded, and a replacement client completed after one client disconnected. Per-client p95 latency ranged from 16.09 to 16.46 seconds because the single-threaded daemon serialized saturated inference.

The high latency is retained as a systems result, not averaged away. The run demonstrates response correlation, fairness, isolation, and recovery in one serial guest. It does not demonstrate parallel model execution, distributed consensus, field capacity, or multi-host availability. The action execution method was unavailable on these sockets with JSON-RPC error -32601. Execution records, physical delivery attempts, and physical deliveries were all zero.

7.3 Visible natural-use sessions

Three independently booted disposable sessions were operated through the visible FerrumOS assistant using Windows computer control. A frozen prompt set produced 24 privacy-bounded telemetry rows across six action classes. Eighteen read-only actions executed successfully, three write actions remained awaiting operator confirmation, and three delete actions were blocked by the gate.

The committed telemetry excludes prompts, arguments, paths, provider identifiers, model identifiers, and output text. Language-model page-ins after deterministic requests, guest faults, synthetic collector markers, direct execution RPCs, physical deliveries, and source-disk changes were zero. This is short researcher-operated QEMU evidence. It is not a seven-day study, independent user testing, production telemetry, or a labelled precision/recall dataset.

7.4 What the OS evidence supports

Together, the OS experiments support a narrow systems claim: a learned preview can run inside the real unprivileged mediation path while execution authority remains separately unavailable or gated. The experiments also expose a practical bottleneck under saturated serial inference. They do not rescue the zero learned-contribution result in Section 6. The strongest learned result remains architecture-controlled rollout prediction; the strongest authority result remains deterministic separation and absence of unauthorized effects.

8. Physical evidence and retained negatives

8.1 Externally authored case intake

The external intake freezes Microsoft Azure VM-noise revision `207bed67dd10090b28ad4f745b2cfd41a11aace4` as an OS workload taxonomy and NVIDIA Anchor-Lab revision `647edd5787cd764cdc041103ad282dc59214d919` as physical telemetry. Azure families cover file, process, thread, program launch, CPU, memory, random I/O, and Redis contention. They inform the natural-use and contention mix but are not projected into FerrumOS vectors or treated as FerrumOS labels.

Six Anchor-Lab Parquet files contain 2,925,558 finite timestamped rows, including three publisher-designated held-out SO-101 trials and three H1 elbow-teststand conditions. The files contain command or target and measured-state telemetry but no safety or contact labels. Their joint, actuator, and temperature semantics do not match Physical JEPA v5's 16-state navigation and maintenance representation or seven-action ontology. No projection, inference, or actuator delivery was performed. The incompatibility is a useful negative intake result: it prevents a large external row count from being misrepresented as direct model validation.

8.2 Recorded HAI sensor replay

The frozen Physical JEPA v5 artifact is evaluated over 284,398 one-second transitions from HAI 23.05 test1 and test2 using previously frozen projection statistics. Six conditions apply clean replay, two-second latency, zero-to-three-second jitter, 0.02 sensor noise, 5% hold-last dropout, and a combined fault with proxy-command feature saturation.

Condition	H=1 MAE	Delta	AUROC	Event recall
Clean	0.046627	0.000000	0.6115	38.46%
Latency 2 s	0.050421	0.003794	0.6040	48.08%
Jitter 0-3 s	0.049488	0.002861	0.6039	44.23%
Noise 0.02	0.055194	0.008567	0.5937	46.15%
Dropout 5%	0.046735	0.000108	0.6105	38.46%
Combined	0.059456	0.012829	0.6043	50.00%

Host NumPy batch inference measured 2.239 microseconds median and 3.303 microseconds p99 per row. These are host software measurements, not physical control-loop timing. Across 472 injected observation-fault windows, the first clean row after the injection ended fell below 1.25 times clean p95 error. That zero-second observation restoration is not a physical recovery-time claim. Actuator delivery attempts and deliveries were both zero.

The replay is the strongest external physical evidence directly compatible with the present v5 representation, but it remains researcher-executed replay. It does not include live Ferrum hardware, physical clocks and interfaces, actuator dynamics, physical contact recovery, or an independently tested emergency-stop path.

8.3 Multi-embodiment 3D stress

A separate PyBullet DIRECT protocol varies three bodies, three obstacle geometries, mass, three-dimensional targets, contact, and a one-second return-to-start recovery. All 288 cases are stopped by the union policy. Task completion is 0%, with Wilson 95% upper bound 1.32%. The unshielded arm contacts an obstacle in 205 of 288 cases. The union arm records no contacts only because it intervenes in 288 of 288 cases.

The learned branch contributes 23 interventions not produced by the rule; 13 coincide with an unshielded contact. Those cases cannot be credited as useful learned collision avoidance because the policy has no selectivity and completes no task. Simulated return-to-start recovery succeeds in 181 of 205 contact cases, or 88.29%, with Wilson interval [83.17%, 92.01%]. This locally designed software-physics test adds geometry and contact coverage while failing the completion-versus-intervention objective.

3D outcome	Count	Rate or interval
Union interventions	288 / 288	100.00%
Completed tasks	0 / 288	0%; upper 95% 1.32%
Unshielded contacts	205 / 288	71.18%
Learned-only interventions	23 / 288	7.99%
Learned-only and unshielded contact	13 / 288	4.51%
Simulated recovery	181 / 205	88.29% [83.17%, 92.01%]

No shielded contact in this run should be read as a zero underlying collision probability. More fundamentally, avoiding contact by stopping every case is not useful autonomy. Retaining the negative prevents an embodiment-diversity checkbox from being mistaken for deployment progress.

9. Cross-domain synthesis

9.1 Prediction quality is domain-specific

The matched study rejects the simplest architecture story. Physical dynamics strongly favor the JEPA in this registered setting. FerrumOS short-horizon prediction favors the GRU, with the JEPA best only at H=5. The result is more useful than a universal-winner claim because it exposes where architecture selection must remain empirical. It also gives the Physical JEPA paper a controlled dynamics comparison while making the FerrumOS systems claim less dependent on a JEPA label.

9.2 Causal response is not calibrated authority

Both selected models react to paired interventions in the intended direction. Neither generates an operational intervention at the conservative frozen threshold on the shifted final catalogs. These findings are compatible: a model can encode direction and approximate magnitude while its calibrated score lacks separation at a chosen false-positive constraint. Reporting only rollout or directional accuracy would conceal the actual decision failure.

9.3 Deterministic authority can fail differently

The new 512-case catalogs show no interventions from either rules or learning. The 3D stress shows intervention on every case. One extreme has no hazard recall; the other has no task completion. Together they show why a shield should be evaluated on both avoided harm and intervention cost at a registered operating point. Deterministic authority is necessary as a control boundary but is not automatically sufficient as an engineering policy.

9.4 Evidence ladders prevent category errors

QEMU timing is not production throughput. Researcher-operated sessions are not independent user evidence. Recorded sensors are not live HIL. PyBullet contact is not physical collision evidence. A publisher-heldout robotics file is not compatible merely because it contains robot telemetry. Labelling those boundaries does not weaken the contribution; it makes the evidence composable and reproducible.

9.5 Novelty statement

This work does not claim the first JEPA world model, first safety filter, first calibrated predictor, or first runtime-assurance architecture. Its contribution is the cross-domain combination of: matched architecture control; explicit separation of prediction, policy, capability, and effect; sealed validation-only selection; paired causal and operational estimands; runtime evidence with authority disabled; semantic compatibility checks for external data; retained operational negatives; and digest-verified non-promotion. The principal hypothesis supported is methodological: predictive evidence and authority evidence should be registered, measured, and reviewed separately.

10. Threats to validity and limitations

1. The final temporal catalogs and 3D benchmark use deterministic simulator labels designed locally. They are blinded after registration but not independently designed or assessed.
2. The architecture result is controlled within the chosen data, state representations, parameter scale, optimizers, and update budget. It does not establish universal superiority for any model family.
3. Calibration is weak under the registered temporal shift. ECE is bin-dependent, and risk-coverage ordering is not monotonic. No calibrated safety guarantee is claimed.
4. The operational zero-result is threshold-specific. Lower thresholds could increase recall and false positives, but selecting one after final access would invalidate the frozen operating-point claim.
5. FerrumOS runtime evidence uses QEMU/WHPX, disposable disks, a serial daemon, four local clients, and 24 requests from one researcher. There is no production load, multi-host field test, independent user study, or long-duration natural-use telemetry.
6. The Physical JEPA replay uses an explicit projection of recorded testbed sensor streams. There is no live Ferrum HIL, physical actuation, physical timing, sensor-interface latency, actuator dynamics, human contact, or hardware emergency-stop validation.
7. The HAI replay's event labels and proxy features do not convert it into an end-to-end safety benchmark. Host microsecond measurements and one-row observation restoration are not control-loop or physical-recovery timings.
8. Anchor-Lab telemetry is externally authored but semantically incompatible with direct v5 scoring. It supports a future embodiment-specific model, not transfer of the present one.
9. The PyBullet environment is locally designed and simple relative to robotics benchmarks. Its 100% intervention rate and 0% completion make it a negative stress test, not evidence of practical collision avoidance.
10. Deterministic rules are engineering predicates, not formally verified invariants. The empirical absence of an effect does not prove that every execution path is impossible.
11. No protected research result was promoted. The report therefore evaluates a research lineage, not a deployed policy change.

11. Reproducibility and artifact integrity

The umbrella verifier binds protocols, selections, results, runtime reports, external intake, replay amendments, stress tests, and subordinate verifiers by SHA-256. It independently recomputes protected deployment digests and requires `promotion_eligible=false`. Selection attempts are logged, final paths are denied during model and threshold choice, and once-opened results write to new files.

Principal reproduction commands are:

```
python scripts/verify_cross_domain_world_models.py
python scripts/verify_cross_domain_learned_contribution.py
node scripts/verify_world_model_runtime_benchmark.mjs docs/research/world_model_v3_4_shadow_runtime_v1.json
python scripts/verify_world_model_v3_4_shadow.py
python scripts/verify_world_model_multiclient_result.py
python scripts/verify_world_model_natural_use.py
python scripts/verify_world_model_external_case_intake.py
python scripts/verify_physical_jepa_recorded_hil_replay.py
python scripts/verify_physical_jepa_multi_embodiment_3d.py
python scripts/verify_cross_domain_world_model_improvement_study.py
python scripts/verify_cross_domain_world_model_paper.py
```

The paper verifier checks required claims and boundaries in the manuscript, PDF text and metadata, figure presence, page count, source and PDF digests, the umbrella verification result, non-promotion status, and protected-artifact integrity. A paper freeze record identifies the exact manuscript, rendered PDF, figures, and evidence snapshot used for Technical Report v1.0.

12. Conclusion

This report finds a real architecture-controlled advantage for Physical JEPA and a different ranking in FerrumOS. It also finds that neither selected learned model adds a single intervention at the frozen conservative operating point. The 3D stress test fails in the opposite direction by stopping every task. Those are not contradictory results. They locate three different engineering problems: learning dynamics, calibrating decisions under shift, and designing authority policies that preserve both safety and useful completion.

The defensible contribution is therefore not that a world model makes an operating system or robot safe. It is that world models can be evaluated inside a reproducible authority architecture without being mistaken for authority. Prediction remains advisory; deterministic policy, capability, confirmation, and actuator denial remain independently enforceable; negative evidence remains visible; and deployment remains unchanged unless every frozen gate supports promotion. In this study, they do not.

12.1 Submission scope

The appropriate submission identity is a cyber-physical systems, runtime-assurance, or dependable-agent-systems paper. The Physical JEPA experiments include externally recorded sensors and software physics, but no live actuation; calling the work a robotics-deployment paper would exceed the evidence. The FerrumOS experiments include a real in-guest mediation path but no production population. The cross-domain value is the shared authority method and the controlled contrast, not an assertion that the two environments are equivalent.

12.2 Highest-value next evidence

The next study should not add another locally designed benchmark merely to increase case count. It should target the two observed decision failures: no recall at the conservative shifted threshold and no completion under the 3D all-stop policy. A blinded benchmark should freeze a useful-operating-region objective before data access, jointly requiring materially lower harmful outcomes, high task completion, and low intervention. Calibration should be fitted without final access and reported with uncertainty. For the physical lineage, actuator-disabled live HIL with measured sensor latency, actuator-interface timing, independent emergency stop, and recorded physical clocks is the shortest path to a stronger evidence class. For FerrumOS, a longer independently operated natural-use study and a concurrent inference implementation are higher value than additional synthetic prompts.

Priority	Frozen success criterion	Why it changes the evidence tier
Blinded useful-autonomy benchmark	High completion, lower harmful outcomes, low intervention	Replaces the zero-recall/all-stop extremes with a joint operating objective
Actuator-disabled live HIL	Physical clocks and interfaces observed; authority remains zero	Adds live integration without claiming autonomous actuation
Independent execution and assessment	Protocol run and labels controlled outside the author workflow	Reduces local-design and researcher-operation bias
Calibration under shift	Pre-registered threshold with reliability and uncertainty intervals	Tests whether predictive skill becomes stable decision value
FerrumOS concurrent preview	Bounded latency under independent clients with no leakage	Addresses the measured serial contention bottleneck

12.3 Release rule

Future improvements should create new versioned research artifacts and preserve every failed frozen result. A candidate may be promoted only under a separate prospective deployment protocol that identifies the exact protected targets, rollback path, capability boundary, post-execution verification, and all required gates. Publication of this report does not satisfy that protocol.

Appendix A. Claim-to-evidence ledger

This ledger is part of the report rather than a supplementary marketing summary. It identifies the evidence object that supports each major statement and the closest claim that the same object does not support. The purpose is to make category errors visible during review.

Claim	Primary evidence	Supported result	Explicit exclusion
Physical JEPA leads matched alternatives	Shared frozen physical final catalog	Lowest H=1, H=3, H=5 rollout error; paired intervals exclude zero	Not universal JEPA superiority
FerrumOS ranking differs	Shared frozen FerrumOS final catalog	GRU leads H=1 and H=3; JEPA leads H=5	Not evidence that recurrent models are universally best
Selected models encode interventions	Paired temporal counterfactual catalogs	100.00% and 93.36% directional accuracy	Not calibrated hazard recall
Shifted scores are weakly calibrated	Frozen reliability, Brier, and ECE outputs	ECE 0.238143 and 0.292887	Not a formal probability guarantee
Learned models add operational caution	Three-arm 512-case catalogs	Zero marginal blocks at threshold 0.99	No learned safety-value claim
Runtime preview is integrated	FerrumOS QEMU ring-3 shadow path	H=1-H=5 p99 guest time 3 ms	Not production or hard real-time timing
Multiple clients remain correlated and isolated	Four WebSocket clients, 128 responses	No leakage; Jain fairness 0.999937	Not parallel inference or distributed execution
Visible assistant mediation works in bounded sessions	Three QEMU boots, 24 requests	Reads execute; writes await confirmation; deletes block	Not independent users or production telemetry
External physical streams can be replayed	284,398 HAI transitions	Fault-condition error and event diagnostics	Not live Ferrum HIL or physical recovery
Anchor-Lab adds external embodiment data	Six publisher files, 2,925,558 rows	Timing and command/state fields are finite	Not semantically valid for direct v5 scoring
3D geometry/contact stress is exercised	288 local PyBullet DIRECT cases	Contact and simulated recovery are measured	Not practical learned safety at 100% intervention
Deployment stayed unchanged	Recomputed protected SHA-256 digests	Every protected artifact is byte-identical	Not a deployment or release result

A.1 Evidence precedence

When two measurements appear to support different narratives, the operationally closer measurement takes precedence for the operational claim. For example, the physical JEPA's rollout advantage remains valid even though its learned-only intervention count is zero. The former supports dynamics prediction; the latter controls any statement about safety-gate value at threshold 0.99. Likewise, zero union contacts in the 3D run does not supersede 0% task completion and 100% intervention. The complete outcome vector is the result.

A.2 Negative-result taxonomy

The study distinguishes three types of negative evidence. A scientific negative is a completed comparison whose registered estimand does not support the hoped-for effect, as in zero learned marginal caution. An engineering negative is a system behavior that is measurable but unusable, as in serial multi-client latency or the all-stop 3D policy. A compatibility negative occurs before model scoring when external data do not share the required semantics, as with Anchor-Lab. None is a failed run, and none should be rewritten as missing data.

Appendix B. Frozen-gate and artifact audit

The paper inherits the study's frozen-gate discipline. Each verifier checks its own source objects and writes a result that is subsequently bound by the umbrella verification record. The paper verifier does not recompute every experiment; it requires the already committed umbrella result to pass, verifies its non-promotion and protected-digest assertions, and then binds the manuscript and rendered PDF to that evidence snapshot.

Stage	Frozen before final access	Verification requirement	Mutation boundary
Architecture selection	Methods, seeds, updates, parameter tolerance, validation rule	Final inaccessible during every selection run	New research model outputs only
Architecture evaluation	Selected checkpoints and final-catalog generator	One final opening; paired bootstrap finite	No deployed artifact path
Learned contribution	Model, calibrator, threshold, three arms	One final opening; honest zero marginal result	No post-final threshold tuning
FerrumOS shadow	Guest image source, command classes, horizons	Correlated responses, runtime fields, source-disk hash	Disposable disk copies only
Multi-client contention	Four-client schedule and correlation IDs	128 responses, isolation, fairness, recovery	Execution RPC unavailable
Natural use	Prompts, sessions, privacy schema	24 bounded rows; forbidden fields absent	Researcher-operated disposable guests
External intake	Source revisions and semantic checklist	File counts, finite rows, compatibility decision	No invalid feature projection
Recorded replay	Projection, faults, descriptive threshold	All transitions finite; authority counters zero	No artifact retraining or actuation
3D stress	Bodies, obstacles, cases, recovery rule	All outcomes and Wilson intervals retained	PyBullet DIRECT; actuator authority zero
Paper freeze	Required claims, boundaries, figures, metadata	Text, hashes, pages, evidence snapshot pass	Documentation artifacts only

B.1 Protected deployment inventory

The umbrella record protects four deployed objects: the FerrumOS encoder, FerrumOS transition, FerrumOS manifest, and Physical JEPA transition. Their observed digests equal their registered expected digests. The cross-domain architecture models, calibration objects, final catalogs, replay projections, and 3D results are research artifacts. A passing paper verifier cannot change that status. Promotion would require a separate prospective protocol and every deployment-specific gate.

B.2 Reproduction order

A reviewer can begin with `verify_cross_domain_world_model_improvement_study.py`, which checks the subordinate verification records and protected digests. For deeper inspection, the architecture and learned-contribution verifiers expose selection-access logs and final-open accounting; runtime verifiers expose no-execution and source-disk checks; physical verifiers expose actuator-authority counters and result attribution. The final paper verifier then checks that the manuscript says what the evidence permits, not merely that a PDF exists.

B.3 Archival boundary

Technical Report v1.0 is intended to be archived together with its manuscript, figures, paper verification record, and freeze manifest. Dataset or software records may receive their own archival identifiers because they are independently reusable objects. Reciprocal links should identify the exact immutable report version and the exact dataset or software version without implying that a repository's moving branch is itself frozen. The DOI fields are intentionally not invented in this pre-archive build; they can be added only after the archival service reserves or publishes the corresponding identifiers.

Appendix C. Artifact locator

The repository is the executable supplement to the narrative. Table C.1 lists the shortest path from a headline claim to its primary machine-readable evidence. Verification files are intentionally separate from result files so that a result producer is not the sole authority for its own gate.

Evidence object	Repository path	Review use
Prospective cross-domain protocol	`docs/research/cross_domain_world_model_improvement_protocol_v1.json`	Registered methods, seeds, budgets, access and promotion rules
Architecture selection	`docs/research/cross_domain_world_model_selection_v1.json`	Check chosen family and denied final access
Architecture result	`docs/research/cross_domain_world_model_architecture_result_v1.json`	Recompute H=1, H=3, H=5 tables and paired intervals
Learned-contribution result	`docs/research/cross_domain_learned_contribution_result_v1.json`	Recompute three-arm confusion matrices and marginal counts
FerrumOS shadow result	`docs/research/world_model_v3_4_shadow_runtime_v1.json`	Inspect guest timing, loaded artifact and no-execution fields
Multi-client result	`docs/research/world_model_multiclient_contention_result_v1.json`	Inspect correlation, fairness, leakage and recovery
Natural-use result	`docs/research/world_model_natural_use_result_v1.json`	Inspect bounded sessions and privacy assertions
External intake result	`docs/research/world_model_external_case_intake_result_v1.json`	Inspect source revisions, rows and compatibility decision
Recorded replay result	`docs/research/physical_jepa_recorded_hil_replay_result_v1.json`	Recompute fault-condition and authority-disabled metrics
3D stress result	`docs/research/physical_jepa_multi_embodiment_3d_result_v1.json`	Recompute completion, intervention, contact and recovery
Umbrella verification	`docs/research/cross_domain_world_model_improvement_verification_v1.json`	Confirm subordinate passes, claim boundaries and protected hashes
Narrative study	`docs/research/CROSS_DOMAIN_WORLD_MODEL_IMPROVEMENT_STUDY.md`	Read the compact evidence-first study before this full report

C.1 Recommended audit path

Begin with the prospective protocol and umbrella verification. Confirm `promotion_eligible=false`, all protected `unchanged` fields, and the claim-boundary list. Then inspect the architecture and learned-contribution results together: this prevents a favorable rollout table from being detached from the operational zero-result. Continue to the runtime and physical records only for the integration claims they support. Finally, execute the paper verifier and compare its recorded manuscript and PDF digests with the archived files.

C.2 Versioning rule

The report version identifies a fixed document, not a promise that every moving repository file will remain identical forever. Corrections that alter prose, evidence binding, or layout should create a new immutable report version and retain the old version in archival history. Dataset and software deposits should use their own versioned records when they are independently reusable. A concept DOI may identify the evolving record, while the paper should cite the immutable version DOI used during review.

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